SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Case Study No.: 4

1. Case Study Title: ATM Card Pin Verification using JavaScript Functions Case Study.

2. Case Study Program Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>ATM Card PIN Verification Case Study</title>
    <style>
        body {
            font-family: Arial, sans-serif;
```

```css
    background-color: #f4f6f9;
    margin: 40px;
    display: flex;
    justify-content: center;
}
.container {
    width: 100%;
    max-width: 500px;
    background: white;
    padding: 25px;
    border-radius: 8px;
    box-shadow: 0 4px 10px rgba(0,0,0,0.1);
}
h2 { color: #1e3a8a; margin-top: 0; }
.meta-info {
    background-color: #eff6ff;
```

```
    padding: 10px;

    border-left: 4px solid #3b82f6;

    font-size: 14px;

    margin-bottom: 20px;

}

input[type="text"] {

    width: 100%;

    padding: 12px;

    margin: 10px 0;

    border: 1px solid #cbd5e1;

    border-radius: 4px;

    box-sizing: border-box;

    font-size: 16px;

}

button {

    width: 100%;
```

```css
    background-color: #1d4ed8;

    color: white;

    padding: 12px;

    border: none;

    border-radius: 4px;

    font-size: 16px;

    cursor: pointer;

    font-weight: bold;

}

button:hover { background-color: #1e40af; }

.log-screen {

    margin-top: 20px;

    background-color: #1e293b;

    color: #38bdf8;

    padding: 15px;

    border-radius: 4px;
```

```
            font-family: 'Courier New', Courier, monospace;

            font-size: 13px;

            white-space: pre-line;

            min-height: 80px;

            border: 2px solid #0f172a;

        }

    </style>

</head>

<body>



<div class="container">

    <h2>ATM Card PIN Verification</h2>



    <!-- Student Header Details -->

    <div class="meta-info">

        <strong>Name:</strong> Sonal Nakhate <br>
```

```
        <strong>PRN:</strong> 24070521199 <br>

        <strong>System:</strong> Global Secure Bank ATM

    </div>



    <label for="pinInput"><strong>Enter Card PIN:</strong></label>

    <input type="text" id="pinInput" placeholder="e.g., 1221, 4554, 9821"
maxlength="6">

    <button onclick="runVerificationPipeline()">Verify ATM PIN</button>



    <h3>ATM Terminal Output Logs:</h3>

    <div id="terminalLog" class="log-screen">Awaiting card insertion and
PIN entry...</div>

</div>



<script>

    // ==========================================

    // GLOBAL SCOPE CONFIGURATIONS
```

```
    // ==========================================

    const BANK_NAME = "Global Secure Bank";

    const DAILY_ATTEMPTS_LIMIT = 3;

    let attemptsUsed = 0; // Tracks active transaction global scope
counters



    // ==========================================

    // 1. FUNCTION DECLARATION: Reverses a PIN string

    // ==========================================

    function reversePin(pin) {

        const pinString = String(pin);

        const reversed = pinString.split("").reverse().join("");

        return reversed;

    }



    // ==========================================

    // 2. FUNCTION EXPRESSION: Checks if a PIN is a palindrome
```

```
// ===========================================

const isPalindromePin = function(pin) {

    const pinString = String(pin);

    const reversedPin = reversePin(pinString);

    return pinString === reversedPin;

};



// ===========================================

// 3. ARROW FUNCTION: Formats output alerts based on validation

// ===========================================

const generateSecurityAlert = (pin, isPalindrome) => {

    if (isPalindrome) {

        return `[SPECIAL SECURITY PROTOCOL]: PIN ${pin} is a 
Palindrome.\nHigh-symmetry alert triggered. Proceeding to main menu...`;

    }

    return `[STANDARD SECURITY]: PIN ${pin} verified 
successfully.\nProceeding to main menu...`;
```

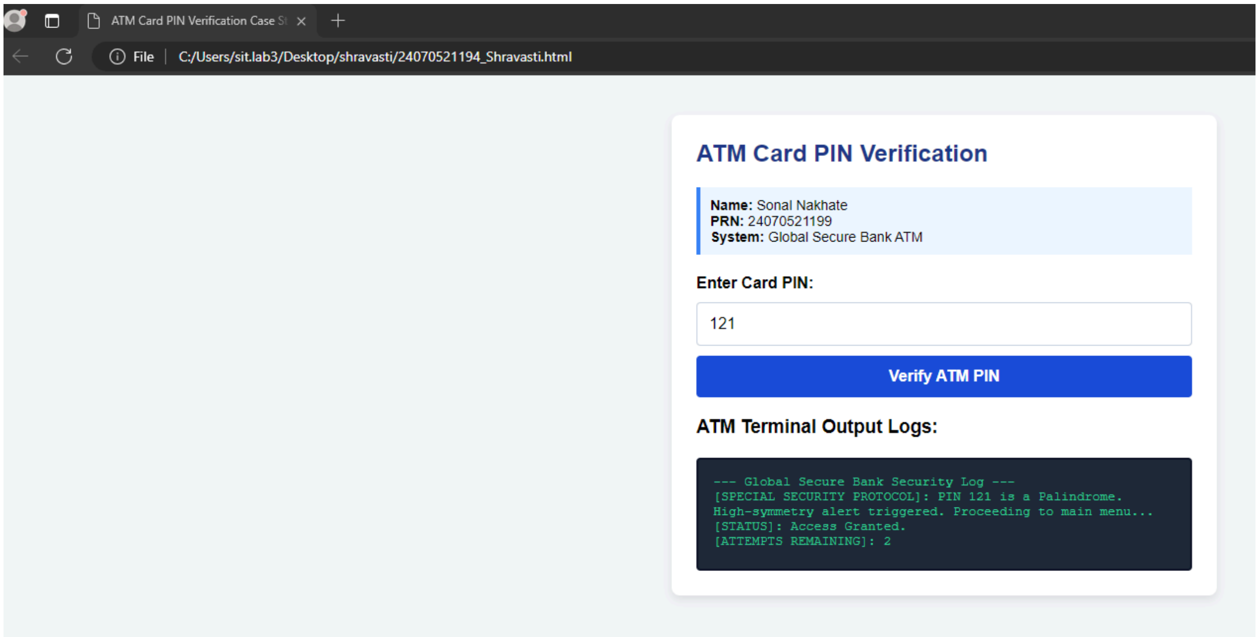
```
};


// ==========================================
// 4. MAIN PROCESS: Scope Management & Try-Catch
// ==========================================
function runVerificationPipeline() {
    // LOCAL SCOPE VARIABLES
    const pinInputField = document.getElementById("pinInput");
    const terminalLog = document.getElementById("terminalLog");
    const enteredPin = pinInputField.value.trim();


    console.log(`--- Processing PIN verification at ${BANK_NAME}
---`);


    try {
        // TRY BLOCK validation checks
        if (enteredPin === "") {
```

```
                throw new Error("Transaction Cancelled: PIN field cannot
be empty.");
            }


            if (isNaN(enteredPin)) {
                throw new Error("Security Alert: Alphabetic characters are
invalid.");
            }


            if (attemptsUsed >= DAILY_ATTEMPTS_LIMIT) {
                throw new Error("Critical Lockout: Account locked.
Exceeded daily limits.");
            }


            // Processing safe operational parameters
            attemptsUsed++;
            const isPalindromic = isPalindromePin(enteredPin);
            const alertMsg = generateSecurityAlert(enteredPin,
isPalindromic);
```

```
            const remainingAttempts = DAILY_ATTEMPTS_LIMIT - attemptsUsed;



            // Generate clean textual UI output profiles

            const successReport = `--- ${BANK_NAME} Security Log ---

            ${alertMsg}

            [STATUS]: Access Granted.

            [ATTEMPTS REMAINING]: ${remainingAttempts}`;



            terminalLog.innerText = successReport;

            terminalLog.style.color = "#34d399"; // Green tint on success



            console.log(`PIN: ${enteredPin} Verified. Palindrome Status:
${isPalindromic}`);



        } catch (error) {

            // CATCH BLOCK for graceful exception containment

            const errorReport = `--- ${BANK_NAME} SYSTEM FAULT ---
```

```
            [ERROR]: ${error.message}

            [STATUS]: Transaction Denied.

            [TOTAL ATTEMPTS USED]:
${attemptsUsed}/${DAILY_ATTEMPTS_LIMIT}`;



            terminalLog.innerText = errorReport;

            terminalLog.style.color = "#f87171"; // Red tint on errors



            console.error(`[SYSTEM TRACE FAULT]: ${error.message}`);

        }

    }

</script>



</body>

</html>
```

3. Output:



4. Result/Conclusion: We have successfully implemented a simple JavaScript program that checks if the ATM pin is a palindrom or not.